attacks or failures of individual nodes. Decentralisation also enhances trust and transparency. No single party controls the data in decentralised consensus techniques, and the entire procedure is open and verifiable.

Security: Data integrity is preserved and the network is protected against malevolent attacks. To maintain data integrity and safe communication, consensus systems mostly rely on cryptographic techniques.

Fault tolerance: This is a crucial component of consensus algorithms in blockchain systems, and makes sure the network keeps running even if some nodes are hostile or malfunctioning. Achieving consensus in a decentralised setting necessitates strong fault tolerance techniques because nodes may malfunction, act maliciously, or behave inconsistently.

Consistencies: All trustworthy nodes in a distributed network concur on the same blockchain state. The correctness and integrity of the data being transferred throughout the network are ensured by this essential component of blockchain systems.

Byzantine Fault Tolerance (BFT)

This consensus method can reach a consensus in a distributed system even if up to a third of the nodes malfunction or behave maliciously. It deals with the Byzantine Generals Problem, in which nodes must agree even while some of them are giving false or inaccurate information. Nodes (or replicas) in BFT systems communicate with one another to confirm transactions and reach a consensus over the state of the system. Even if some nodes are hacked or flawed, more than two-thirds of the nodes must concur on the same outcome in order to agree. Practical Byzantine Fault Tolerance (PBFT), one of the practical applications of BFT, improves efficiency by lowering the required number of communication rounds.

BFT's fundamental benefits are its great resilience and dependability, which allow it to maintain consensus even in the face of malicious activity or node failures. Because of its resilience, BFT is perfect for crucial applications with low levels of trust, such as permissioned blockchains and financial systems. Since BFT doesn't require as much processing power as Proof of Work (PoW), it is more energy-efficient and appropriate for systems with known or semi-trusted nodes. BFT algorithms are effective for real-time applications because they can execute transactions with low latency and high throughput.

BFT does have certain drawbacks, though, chief among them being scalability. It is not feasible for big, decentralised networks, since the quantity of messages sent between nodes increases quadratically with network size. BFT is more suited for permissioned blockchain settings than open, decentralised ones like Bitcoin because of this constraint. In highly hostile contexts, the assumption that less than one-third of the nodes are compromised presents another difficulty. Furthermore, putting BFT into practice can be challenging, requiring exact synchronisation and communication between nodes. BFT is still a key idea in distributed computing despite these difficulties, and has been modified in several blockchain implementations, like Hyperledger Fabric and Tendermint, to guarantee consensus with a predetermined group of participants.

Proof of Work (PoW)

This is the oldest and most popular consensus algorithm, made popular by Bitcoin. Miners must solve cryptographic challenges to validate transactions and append new blocks to the blockchain. The right to attach the block and earn a reward belongs to the first miner who solves the problem. There are basically six steps in this algorithm. The first step is the transaction pool where users make transactions, which are collected into a memory pool of unconfirmed transactions. Miners choose transactions from this pool to include in a new block. In the second step, the miner creates a block with various parameters. In the third step, a valid hash that meets specific criteria is found, and it starts with a certain number of leading zeroes. If the hash doesn't match, the miner changes the nonce and tries again. In the fourth step, the correct solution is found by trial and error. Once found, the miner broadcasts the new block to the network. In the fifth step, block verification and broadcasting are done. If valid, the block is added to the blockchain and all nodes update copies to accommodate new nodes. In the sixth step, the new block is propagated throughout the network. The miner starts working on the next block using the hash of the current block as the previous hash.

PoW's main advantages are its security and resilience. Because it takes a lot of computing power to solve the cryptographic challenge, it becomes unaffordable for bad actors to change the blockchain, guaranteeing data integrity and decentralisation. Furthermore, regardless of the network's processing capacity, consistent block times are guaranteed by the difficulty adjustment algorithm. Because of this feature, PoW is extremely resilient to Sybil attacks, in which one party attempts to take control of the network.

However, PoW has quite a few problems, the most important being the energy it uses. Because only companies with substantial financial resources can compete, mining necessitates specialised gear (such as ASICs), raising worries about centralisation. There are environmental issues too, since the processing power used during mining does little more than secure the network. As only one miner eventually receives the reward, the competition among miners results in wasted energy utilisation.